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Chatlaw: A Multi-Agent Legal Assistant based on a Role-Aligned Mixture-of-Experts Architecture

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ABSTRACT

Artificial Intelligence (AI) holds great potential in legal services, yet Large Language Models (LLMs) face two major challenges: limited knowledge of the Chinese legal system and vulnerability to hallucinations. To address these issues, we present Chatlaw, a multi-agent legal assistant. Chatlaw's framework is designed to emulate the Standard Operating Procedures (SOP) of real law firms, where different roles (e.g., assistant, researcher, senior lawyer) collaborate on a case. To computationally mirror this collaborative structure, we developed a novel Role-Aligned Mixture-of-Experts (RA-MoE) architecture. In this system, the internal "experts" are specifically trained to align with the distinct tasks of each agent role (e.g., inquiry, analysis, drafting). These specialized agents (Legal Assistant, Researcher, etc.) then form the collaborative framework. When they interact with users, retrieve legal knowledge, analyze case details, or generate reliable consultations, the RA-MoE architecture intelligently routes their computations to the corresponding dedicated expert, ensuring each step is handled by the most qualified parameters. In evaluations, Chatlaw surpasses general-purpose AI models, including GPT-4, achieving a 7.73% improvement in accuracy on the LawBench benchmark and an 11-point higher score on the Unified Qualification Exam for Legal Professionals. Real-case studies and expert assessments further confirm its robustness. Chatlaw enhances the accessibility and reliability of legal services, advancing the provision of legal support to the public.

Keywords: artificial intelligence in law; mixture of experts; large language model; knowledge graph; legal assistant; legal technology

1. Introduction

Legal services play a crucial role in protecting individual rights and maintaining social fairness [1–3]. However, the limited availability of legal professionals and the high cost of their services often restrict access to these services, particularly in China, with its vast population and extensive social interactions. This gap in legal service provision deeply impacts justice and equity, especially for those lacking the resources to effectively navigate the legal system. Besides the widespread demand for legal services, China’s legal system has two significant characteristics: First, unlike the United States, where each state has different legal regulations, China’s legal system is unified. Second, China’s centralized legal system and extensive legal infrastructure provide a unique and suitable environment for the research and development of legal AI. Therefore, this raises a crucial question: can we establish an automated legal assistant to address the specific needs of one of the world’s largest and most uniform legal systems?

In recent years, the efficacy of Large Language Models (LLMs) has been validated across multiple scientific fields, encompassing natural language processing [4,5], human interaction [6,7], biochemistry [8–12], and the medical field [13–20]. LLMs also offer potential solutions to the challenges in legal services. Popular models like ChatGPT [21] and the LLaMA [22] series, along with other general-purpose [23–27] or law-specific models [28–30], can respond to user inputs based on their internal legal knowledge repositories and provide advisory recommendations. However, the inherent hallucination issues in LLMs pose potential risks in their application to legal domains [31] since they operate at the level of word distributions rather than validated facts [32]. The knowledge generated by these models is often incomplete or outdated, leading them to produce illusions that, although seemingly relevant, may be misleading or incorrect [33].

To address these limitations, we propose Chatlaw, a multi-agent virtual legal assistant based on a novel architecture designed to align computational specialization with real-world legal workflows. The core innovation lies in the Role-Aligned Mixture-of-Experts (RA-MoE) architecture. Rather than using a single LLM for all tasks, we leverage a task-specific Mixture-of-Experts (MoE) model [34,35], where each "expert" is specialized for specific tasks based on the Standard Operating Procedures (SOPs) of legal service workflows. SOPs in law firms guide task delegation (e.g., information gathering, research, legal advice, report writing), and this framework ensures that each agent’s role is matched with an optimal, task-specific model [36].

This alignment of experts to specific legal functions mitigates hallucination risks and enhances both efficiency and accuracy. For example, a model optimized for legal research may not be ideal for generating legal advice or drafting documents. By tailoring each agent to its function, we improve both task-specific performance and overall workflow efficiency. Additionally, we support this system with a high-quality legal dataset and knowledge graph, ensuring the accuracy of the information provided by each agent.

Our model demonstrates superior performance in various evaluations, surpassing

existing large language models, including GPT-4 [21]. Specifically, it outperforms GPT-4 in the Lawbench and Unified Qualification Exam for Legal Professionals by 7.73% in accuracy and 11 points, respectively, and receives the highest scores in real-case evaluation feedback from legal experts on 4 dimensions, completeness, correctness, guidance, and authority.

2. Related Work

2.1. Mixture-of-Experts Architecture

MoE is a powerful technique used to achieve computational efficiency in large models. The core idea of MoE is to replace dense feed-forward networks with a sparse mixture of "expert" layers, where only a subset of experts is activated during each inference step. This allows models to scale more effectively by significantly reducing the computational cost, especially in terms of floating-point operations (FLOPs). Early work on MoE includes GShard [37], which demonstrated the scalability of MoE to large-scale systems, and Switch Transformers [38], which improved performance while reducing computational redundancy by activating a sparse subset of experts. More recently, open-source models like DeepSeek [39] and Mixtral 8x7B [40] have demonstrated the potential of MoE in delivering state-of-the-art performance in natural language processing tasks.

Despite its successes, traditional MoE architectures suffer from a lack of task-specific specialization. In standard MoE models, the router learns to assign tokens to experts based on the distribution of data [41], without aligning these assignments with any specific high-level task such as reasoning or drafting. This general-purpose specialization limits MoE's ability to handle domain-specific tasks where expert alignment with specific roles is essential. To address this limitation, our work introduces a RA-MoE architecture, where the router is explicitly trained to align expert selections with high-level agent roles defined in our multi-agent framework. This ensures that the experts activated in each task are functionally specialized, thereby enhancing the system's performance in complex multi-agent workflows.

2.2. Multi-Agent Systems

Multi-agent systems (MAS) [42] have been explored as a means to overcome the limitations of single monolithic models in complex task execution. In these systems, multiple agents with specialized functions collaborate to achieve a common goal. Early frameworks, such as ReAct [43], enabled LLMs to call external tools by combining reasoning and acting capabilities. Building on this, more advanced collaborative frameworks like AutoGen [44] have been developed, where multiple agents with different specializations collaborate to solve complex problems.

However, most existing multi-agent systems focus primarily on the macro-level orchestration of workflows [45,46]. These systems still rely on the same underlying, general-purpose LLM APIs (e.g., GPT-4) for all agents, leading to significant computational redundancy. This issue arises because all agents, regardless of their

specialized roles, are forced to share the same computational resources.

Our work addresses this challenge by not only defining the agent-based SOP at the macro level, but also introducing a novel micro-level architecture, the RA-MoE. This architecture ensures that experts are specifically aligned with the roles of the agents, thereby improving computational efficiency and reducing redundancy in multi-agent systems.

2.3. *AI in Law*

The application of artificial intelligence in the legal domain has a long history, with early work focusing on legal information retrieval and rule-based expert systems [47,48]. These systems aimed to automate basic legal tasks such as legal research and document classification. The advent of LLMs has significantly advanced AI in Law, with recent efforts focusing on enhancing the depth of legal knowledge and reasoning capabilities of AI systems.

Legal benchmarks such as LawBench [49] and LexGLUE [50] have been developed to evaluate and improve the performance of LLMs in the legal domain. In addition, domain-specific large models like Lawyer LLaMA [30], LawGPT [51] and Fuzi-Mingcha [52] have been created by fine-tuning general-purpose LLMs on legal corpora. While these models have achieved impressive results, they still face significant challenges when handling complex legal consultations. These challenges include functional generalization, where a single model is expected to proficiently handle diverse tasks such as legal retrieval, reasoning, and drafting, and reliability, where the monolithic nature of a single model makes it prone to hallucinations in long-chain tasks [53]. Our work aims to address these challenges by constructing a multi-agent framework that mirrors the SOP of a law firm, where different roles collaborate to ensure the accuracy and consistency of legal outputs.

3. Chatlaw Framework

3.1. *Role-Aligned Mixture-of-Experts*

In this work, we introduce the RA-MoE framework, which extends traditional MoE models by introducing role-conditioned routing and knowledge-aware gating. Traditional MoE architectures, while efficient in terms of computational cost, typically rely on sparse expert activation without considering the semantic diversity of tasks or the context of knowledge required for specific tasks. In contrast, RA-MoE strategically aligns expert activation with the role-specific tasks in legal workflows and integrates domain-specific knowledge into the routing mechanism to reduce errors and hallucinations during legal consultations.

3.1.1. *Role-Conditioned Routing*

A key innovation of RA-MoE is the introduction of role-conditioned routing. In traditional MoE models, expert selection is based purely on token-level representations, which are agnostic to the task or the role performing the task. However, in the context

of legal services, different stages of legal consultation require distinct expert specializations. For instance, legal research, document drafting, and high-level legal reasoning each demand different parameter sets. In RA-MoE, the router input consists of three components: (i) the token embedding x , representing the content of the input text, (ii) the role embedding E_a , which represents the current agent's role in the workflow (e.g., Legal Assistant, Legal Researcher, Legal Editor, Senior Lawyer), and (iii) the knowledge-graph embedding E_k , encoding the relevant legal entities and relationships extracted from a structured legal knowledge base.

The routing distribution over K experts is computed as:

$$G(x, E_a, E_k; \tau) = \text{softmax}\left(\frac{W_g[x; E_a; E_k]}{\tau}\right),$$

where τ controls the sharpness of the distribution. The top K experts are selected based on the highest probabilities, and their outputs are aggregated as a weighted sum:

$$\text{RA-MoE}(x, E_a, E_k) = \sum_{i=1}^K G_i(x, E_a, E_k) \mathcal{E}_i(x),$$

where $\mathcal{E}_i$ denotes the i -th expert's feed-forward sub-network. This role-conditioned routing ensures that the appropriate experts are activated for each legal task, leading to more accurate and interpretable results.

3.1.2. Knowledge-Aware Expert Activation

RA-MoE further incorporates knowledge-aware gating to enhance its ability to handle legal information effectively. The knowledge-graph embedding E_k is crucial for reducing hallucinations and ensuring the reliability of the output. By embedding structured legal knowledge into the routing mechanism, we enable the system to activate experts who are specialized in handling specific legal entities (e.g., laws, cases, statutes) and their relationships. This integration is particularly useful in legal domains where the accuracy of facts is paramount, and hallucinations can lead to legal risks. The knowledge-aware gating allows RA-MoE to filter out irrelevant expert activations and focus on experts that are most relevant to the legal domain context, thus improving overall system reliability.

3.1.3. Agent-Expert Co-Training

In addition to the role-conditioned routing and knowledge-aware gating, RA-MoE introduces agent-expert co-training. Unlike static MoE models where experts are trained independently, RA-MoE allows for shared training between agents and their corresponding expert clusters. During the multi-agent collaboration process, the Legal Assistant, Legal Researcher, Legal Editor, and Senior Lawyer agents leverage expert clusters specific to their tasks, but they also share gradients across tasks to ensure coherence throughout the entire consultation process. This ensures that the system

operates efficiently, with each agent invoking experts dynamically based on the legal task at hand, while maintaining the overall integrity of the legal consultation pipeline.

The co-training process is guided by a task-expert routing loss function $\mathcal{L}_{route}$, which encourages the router to activate the correct expert subset based on the role-specific task. The total loss function for training is:

$$\mathcal{L}_{total} = \mathcal{L}_{reg} + \alpha \mathcal{L}_{route},$$

where $\mathcal{L}_{reg}$ is the standard cross-entropy loss for autoregressive training, and $\mathcal{L}_{route}$ penalizes deviations between the routing probabilities and the intended expert selection.

The $\mathcal{L}_{route}$ loss function is defined as:

$$\mathcal{L}_{route} = - \sum_{e=1}^E T(e|t) \log G_e(x, E_a, E_k; \tau),$$

where $T(e|t)$ is the target distribution for expert e given task t , $G_e(x, E_a, E_k; \tau)$ is the routing probability output for expert e , and τ is the temperature parameter controlling the sharpness of the distribution. This loss function helps ensure that each task type activates the most appropriate experts, improving both efficiency and accuracy.

Algorithm 1: Generic Agent Template in the SOP Framework

Require: Role identifier r , Expert cluster $\mathcal{E}_r$, Task configuration $\mathcal{P}_r$, Input I_t (from user or previous agent)

Ensure: Structured output O_t

Steps:

1. **Initialization:** Load role embedding $E_a(r)$ and task prompt $\mathcal{P}_r$ for role r
2. **Input Reception:** Receive I_t from user (if first agent) or previous agent output O_{t-1}
3. **Pre-Processing:** Parse I_t into structured context C_t (entities, keywords, graph nodes)
4. **Expert Invocation (via RA-MoE):**

Compute routing distribution $G(x, E_a, E_k) = \text{softmax}(W_g[x; E_a; E_k]/\tau)$

Activate top- K experts $\mathcal{E}_r$ aligned with role r

Generate intermediate output $Y_t = \sum_{i=1}^K G_i(x, E_a, E_k) \mathcal{E}_i(x)$

5. Post-Processing: Validate and refine Y_t through factual verification and formatting

6. Output Transmission: Store refined output O_t in shared memory; pass O_t to the next agent in the SOP sequence

3.2. Chatlaw Agent Design

Built upon the RA-MoE architecture, Chatlaw establishes a multi-agent SOP framework that reproduces the workflow of a real-world law firm. Each agent represents a distinct professional role (including the Legal Assistant, Legal Researcher, Legal Editor, and Senior Lawyer) and operates according to a unified execution template. This structure divides a complex legal consultation into four coordinated stages, where each stage is handled by experts optimized for its specific function.

Within this framework, every agent follows a generic operational template. The template specifies how the agent receives inputs, activates its corresponding expert cluster through the RA-MoE, processes information, and delivers structured outputs to the next stage of the workflow. Through this unified design, the system maintains both functional specialization (achieved by role-aligned experts) and procedural coherence (achieved by standardized task execution). Algorithm 1 outlines the general procedure followed by all agents in the SOP framework.

Following the unified template, the four agents perform complementary functions that collectively realize the complete legal consultation workflow. The Legal Assistant serves as the entry point of the system and mirrors the role of a paralegal or junior associate in a law firm. It receives unstructured user inquiries as input and, through the Pre-Processing stage of the template, extracts entities, temporal information, and key issues to build a structured knowledge graph. This graph acts as a factual foundation that is transmitted to the next stage. The Legal Researcher, corresponding to a legal analyst, then processes this structured graph as its input. Through the Expert Invocation procedure of the template, it activates the retrieval-oriented experts within the RA-MoE to search external databases and statutory repositories for relevant laws, judicial interpretations, and precedent cases. The output of this stage is a curated corpus of verified legal materials that grounds the subsequent reasoning process.

The following stage is handled by the Senior Lawyer, who represents an experienced legal advisor. The Senior Lawyer takes the verified corpus and the structured knowledge graph as input and performs high-level legal reasoning using the advisory experts of the RA-MoE. It validates the document's factual integrity and provides the final professional opinion of the SOP pipeline. Finally, the Legal Editor takes the Senior Lawyer's output as input and engages the document-generation experts to compose a well-structured draft of the legal report or consultation document. Through the Post-Processing phase of the template, the Legal Editor ensures stylistic consistency and formal compliance before passing the draft onward to the user. Together, these four agents embody the modular workflow of a real law firm, where each stage corresponds to a distinct professional responsibility and collectively ensures

that the consultation is both accurate and procedurally coherent.

Table 1 summarizes the configuration of each agent and its corresponding expert cluster, along with the core responsibilities and Input/Output behaviors.

Table 1: Role–Expert Mapping and Functional Responsibilities in the SOP Framework

Agent Role	Expert Cluster	Primary Function	Input / Output
Legal Assistant	$\mathcal{E}_{1..2}$	Entity extraction and graph building	Input: query → Output: structured graph
Legal Researcher	$\mathcal{E}_{3..4}$	Retrieval of laws and cases	Input: graph → Output: corpus
Senior Lawyer	$\mathcal{E}_{5..6}$	Legal reasoning and validation	Input: draft → Output: opinion
Legal Editor	$\mathcal{E}_{7..8}$	Drafting and formatting legal docs	Input: corpus → Output: draft

3.3. SOP Integration and Workflow Execution

The complete SOP integrates the four agents into a sequential workflow that mirrors the collaboration pattern of a professional legal team. The process begins with the Legal Assistant, who interacts directly with the user and converts raw inquiries into a structured representation through knowledge-graph construction. This structured information is then transferred to the Legal Researcher, which performs retrieval and evidence gathering, grounding the subsequent reasoning on verifiable legal sources. The refined corpus is passed to the Legal Editor, which consolidates the retrieved knowledge and user context into standardized legal documents such as consultation reports or petitions. Finally, the Senior Lawyer reviews the draft, validates the legal reasoning, and formulates the authoritative opinion that completes the consultation cycle.

Throughout the workflow, all intermediate results are stored in a shared latent memory that maintains continuity across stages. When each agent executes its task, the RA-MoE router dynamically activates the expert cluster aligned with the agent’s role, allowing specialized computation while preserving global context. This design ensures that information flows transparently between agents without redundancy or loss, enabling the system to emulate the procedural rigor of an actual law firm, where assistants prepare data, researchers verify facts, editors formalize content, and senior lawyers deliver the final decision. By coupling RA-MoE-based specialization with SOP-driven coordination, Chatlaw achieves an interpretable and reliable framework for multi-stage legal reasoning and documentation.

Table 2: Legal Data Used for Training the Large Legal Model

Sub-Category	Sample Size	Scope of Legal Coverage	Temporal Validity
Statutes			
Legal provisions	23k	Civil, Criminal, Admin, etc.	Updated to 2025.11
Legal interpretations	71k	SPC & SPP Interpretations	Updated to 2025.11
Regulations	35k	National & Local Regulations	Updated to 2025.11
Cleaned Cases			
Precedents	796k	National Court Hierarchy	2000–2025
Typical cases	59k	SPC Guiding & Typical Cases	2010–2025
Cases of public interest	665k	Civil & Admin Public Interests	2015–2025
Legal AI Competition Data			
LAIC	768k	Legal Knowledge Reasoning	2018–2024
CAIL	312k	Judgment & Statute Prediction	2018–2022
Other Competitions	220k	Diverse Legal NLP Tasks	2018–2024
Legal Examination			
Examination questions	25k	National Qualification Exam	2018–2024
Examination answers	25k	Official & Model Answers	2018–2024
Examination analysis	38k	Expert Reasoning & Analysis	2018–2024
Consultation + Legal Q&A			
Lawyer consultation	73k	Professional Case Consultations	2020–2025
Public consultation	242k	Community Legal Queries	2020–2025
Legal Q&A	42k	General Legal Knowledge Base	2020–2025

Table 3: Task Data Used for Training the Large Legal Model

Sub-Category	Sample Size	Improving Agents
Text Processing		

Sub-Category	Sample Size	Improving Agents
Keyword extraction	52k	Legal Researcher
Text classification	73k	Legal Assistant
Text generation	70k	Legal Editor
Issue Topic Identification	18k	Legal Assistant
Reading Comprehension	29k	Senior Lawyer
Data Annotation		
Named Entity Recognition	24k	Legal Assistant
Relationship extraction	18k	Legal Assistant
Intent recognition	21k	Legal Assistant
Dispute Focus Recognition	21k	Legal Assistant
Argument Mining	22k	Senior Lawyer
Document Processing		
Document proofreading	19k	Legal Editor
Document merging	6k	Legal Editor
Document splitting	32k	Legal Editor
Legal Process Simulation		
Sub-case segmentation	42k	Senior Lawyer
Clarify details	31k	Legal Assistant
Legal document drafting	5k	Legal Editor
Legal argumentation	21k	Senior Lawyer
Event Detection	35k	Legal Assistant
Trigger Word Extraction	28k	Legal Researcher
Statute Prediction	13k	Legal Researcher
Charge Prediction	12k	Senior Lawyer
Sentence Prediction	28k	Senior Lawyer
Simulated Judgment		
Statute matching	120k	Legal Researcher
Sentence calculation	73k	Senior Lawyer
Inference of charges	48k	Senior Lawyer
Criminal Amount Calculation	91k	Senior Lawyer

3.4. Comprehensive Legal Data Pipeline and Preprocessing for Multi-Task Training

To enable Chatlaw to efficiently handle multi-stage legal tasks, we have developed a comprehensive data pipeline that integrates both legal dataset construction and

preprocessing. This pipeline is designed to generate high-quality task-oriented datasets and ensure they are preprocessed into a format suitable for large language model training. As shown in Fig. 1a, the process begins with multi-source data collection, gathering data from statutes, case documents, legal competition datasets, public consultations, and legal examinations. We employ automated tools for deduplication, denoising, and standardization to ensure the data is cleaned and converted into a consistent format, making it suitable for large-scale model training. The result is a diverse legal dataset, as detailed in Table 2, with over 3.4 million samples covering multiple legal domains, with detailed scope of legal coverage and temporal validity.

After the data is collected, we implement a hybrid human-machine curation process to enhance the quality and accuracy of the data. Law students participate in labeling cases and categorizing them, while domain experts define the semantic relationships between legal terms and create structured knowledge graphs. This process ensures the data is not only comprehensive but also legally valid and interpretable, providing the foundation for specialized tasks in the Chatlaw framework. Table 3 shows the task-oriented datasets with specified types of agents associated with each sub-category during the training phase, which include 26 distinct tasks such as keyword extraction, statute prediction, document drafting, and simulated judgment. The training data for RA-MoE is not explicitly divided into subsets for separate agent training; rather, the model is trained on the full dataset as a unified system. These tasks allow Chatlaw to simulate real-world legal processes, supporting multi-agent collaboration across a range of legal applications.

As part of the preprocessing stage, we format the data into standardized instruction-input-output structures compatible with large language models. For instance, legal provisions and judgment documents are transformed into completion tasks, where the system either completes a legal text or fills in missing sections based on a given context. Legal consultation data is formatted into LLaMA-style chat templates, with two task modes: one that includes additional reference information in the input, and another that allows the model to directly quote relevant provisions in the output. This dual-task setup ensures that the model can function effectively in scenarios where reference provisions are either available or absent.

Together, the data pipeline and preprocessing steps enable Chatlaw to effectively utilize both general legal knowledge and task-specific data for training, ensuring the model is capable of performing a wide variety of legal tasks.

4. Results

4.1. Performance Across Legal Benchmarks

To measure the capabilities of our model, we test our MoE model on two benchmarks and compare it with existing models, evaluating the understanding and mastery of legal systems, regulations, case law, and legal procedures, as well as the

ability to apply this knowledge to specific situations.

Table 4: Performance comparison on Lawbench

#	Method	Average Score	Memorization	Understanding	Application
<i>GPT Series</i>					
1	GPT-3.5	42.15	25.93	43.31	44.74
2	GPT-4	52.35	35.29	54.41	54.05
<i>General LLMs</i>					
3	Baichuan2-7B	38.08	27.30	32.42	47.84
4	ChatGLM2-6B	29.88	15.54	23.45	41.50
5	InternLM2-7B	43.78	25.66	41.74	50.87
6	Qwen-7B-Chat	33.82	28.99	32.80	36.31
<i>Legal LLMs</i>					
7	Fuzi-Mingcha	32.08	20.04	30.25	37.39
8	Chatlaw-MoE (ours)	60.08	43.86	62.11	61.60

Table 5: Performance comparison on Unified Qualification Exam for Legal Professionals

#	Method	Average Score	Exam 2018	Exam 2019	Exam 2020	Exam 2021	Exam 2022
<i>GPT Series</i>							
1	GPT-3.5	78	82	68	82	66	92
2	GPT-4	103	102	108	107	81	117
<i>General LLMs</i>							
3	Baichuan2-7B	61	68	69	61	47	56
4	ChatGLM2-6B	34	33	41	41	26	29
5	InternLM2-7B-Chat	41	41	44	51	32	36
6	Qwen-7B-Chat	48	46	61	51	38	43
<i>Legal LLMs</i>							
7	Fuzi-Mingcha	34	40	40	32	28	30
8	Chatlaw-MoE (ours)	115	113	124	143	115	78

4.1.1. Performance on LawBench

We first conduct tests on LawBench [49], a comprehensive evaluation benchmark based on the Chinese legal system. LawBench mainly covers three cognitive levels: 1) Legal Knowledge Memory: Testing the ability of remembering necessary legal concepts, terms, articles, and facts; 2) Legal Knowledge Understanding: assessing whether large language models can understand and interpret entities, events, and relationships in legal texts; 3) Legal Knowledge Application: evaluating the capability for correctly utilizing and reasoning with their legal knowledge to solve different legal tasks in real scenarios.

As shown in Table 4, in terms of average score, our Chatlaw-MoE model significantly outperforms GPT-4 with a score of 60.08 compared to GPT-4’s 52.35. This substantial difference highlights the effectiveness of our model across different cognitive levels. In the memorization category, Chatlaw-MoE leads with a score of 43.86, surpassing GPT-4’s 35.29, demonstrating its superior ability to remember legal concepts and terms. For understanding performance, Chatlaw-MoE achieves an impressive score of 62.11, significantly higher than GPT-4’s 54.41, indicating better comprehension of legal texts. Finally, in the application category, Chatlaw-MoE scores 61.60, outperforming GPT-4’s 54.05, showing its stronger capability in applying legal knowledge to real-world tasks. Among Legal LLMs, Fuzi-Mingcha scores 32.08 on average, with its highest score in understanding at 30.25, while General LLMs see InternLM2-7B as the strongest with an average score of 43.78 and a leading application score of 50.87. Overall, these results underscore Chatlaw-MoE’s superiority in all evaluated metrics, firmly establishing it as the most effective model in the context of LawBench evaluation, and highlighting the significant advancements made in our approach compared to GPT-4 and other models.

4.1.2. Performance on Unified Qualification Exam for Legal Professionals

The other benchmark is the China’s Unified Qualification Exam for Legal Professionals, including single-choice questions, multiple-choice questions, and uncertain-choice questions. These questions cover various legal fields and can effectively assess the understanding and application ability of legal concepts, principles, and provisions for LLMs.

As shown in Table 5, in the unified legal professional exam from 2018 to 2022, our Chatlaw-MoE model consistently outperformed all other models. With an average score of 115, Chatlaw-MoE significantly surpasses GPT-4’s average score of 104. Specifically, Chatlaw-MoE achieved scores of 113, 124, 143, 115, and 78 across the five years, consistently demonstrating superior performance. In comparison, GPT-4’s scores were 102, 108, 82, 82, and 118 respectively. This consistent outperformance highlights Chatlaw-MoE’s enhanced capability in handling legal examination questions, likely due to the multi-expert system design which dynamically selects the most suitable experts for processing based on input features. Among the Legal LLMs, Fuzi-Mingcha had an average score of 34, with its highest performance in 2019 at 40. For General LLMs, Baichuan2-7B was the strongest, with an average score of 61 and its

highest score of 70 in 2019. These results clearly indicate that our Chatlaw-MoE not only outperforms specialized legal models but also excels against general-purpose language models, establishing it as the leading model for legal task performance.

4.2. Hallucination Detection

4.2.1. Computing the hallucination with semantic entropy

To detect hallucinations in Large Language Models (LLMs), we rely on the principle that the model is uncertain about its output when it is arbitrary. One way to measure this uncertainty is by calculating the predictive entropy of the model's output distribution, which quantifies how uncertain the model is about its response. The **predictive entropy (PE)** for an input sentence x is the conditional entropy $H(Y|x)$ of the output random variable Y , which gives the uncertainty of the model's output for a given input. This can be computed as:

$$PE(x) = H(Y|x) = - \sum_y P(y|x) \log P(y|x)$$

where $P(y|x)$ represents the probability of generating output y given input x , and y is one of the possible outputs. A low predictive entropy indicates that the model's output is highly concentrated (more deterministic), while a high predictive entropy indicates a diverse set of possible outputs.

To estimate semantic entropy, we first need to group generated outputs that convey the same meaning, i.e., outputs that are semantically equivalent. This clustering is based on the concept of semantic equivalence, which describes the relationship between two sentences that mean the same thing. Mathematically, we can define the set of all possible sequences of tokens as $S^N \equiv T^N$, where T is the set of tokens and N is the length of the token sequence. The equivalence relation $E(\cdot, \cdot)$ holds between any two sentences that share the same meaning. This equivalence relation leads to the formation of semantic equivalence classes.

Once we have grouped the generated sequences into classes of semantically equivalent outputs, we can estimate the likelihood that a generated sequence belongs to a particular class. Let C represent the set of all equivalence classes, and $P(C|x)$ represent the probability that a sequence belongs to class C . The **semantic entropy (SE)** is then calculated as the entropy over the distribution of these meaning-classes:

$$SE(x) = - \sum_{c \in C} P(c|x) \log P(c|x)$$

This treats the output as a random variable whose event space is the set of meaning-classes. Since we cannot access all possible meaning-classes directly, we estimate the expectation of the entropy using a Monte Carlo integration over the equivalence classes C :

$$SE(x) \approx - \sum_{i=1}^M P(C_i|x) \log P(C_i|x)$$

where $P(C_i|x)$ is the probability of meaning-class C_i for the input x , and M is the number of sampled equivalence classes. This is the standard method for estimating semantic entropy.

In cases where sequence probabilities are unavailable, we can approximate semantic entropy by counting the number of sequences that fall into each equivalence class. This leads to a variant known as discrete semantic entropy, which estimates the distribution over the meaning-classes based on the proportion of generated outputs in each class:

$$P(C_i|x) \approx \frac{M_c}{M}$$

where M_c is the number of outputs in class C_i and M is the total number of sampled outputs. Discrete semantic entropy approximates the underlying distribution using a categorical empirical distribution, and as the number of samples increases, it converges to the standard semantic entropy calculation.

4.2.2. Performance of legal hallucination analysis

We mainly focus on the subset of hallucinations based on “confabulation” [54] where LLMs randomly generate incorrect answers when uncertain, distinguishing this from cases where LLMs are consistently wrong due to erroneous training data [55] or lie to gain human rewards [56]. This type of hallucination often results in answers that seem correct but contain detailed factual errors, due to variations in input details or LLM uncertainty, thereby introducing legal risks.

We use the method of calculating semantic entropy to detect confabulated hallucinations, which estimates the distribution of meanings of generated answers, rather than the entropy distribution of tokens generated by the LLM. Specifically, for each question, we generate multiple possible answers and cluster those with similar meanings together, details of the method can be found in “Computing the hallucination with semantic entropy” of Methods.

In Fig. 2, we demonstrate that Chatlaw significantly outperforms other models in terms of semantic entropy on the Unified Qualification Exam for Legal Professionals. While achieving the highest accuracy, Chatlaw also obtained the lowest Average Entropy (0.4), as well as the highest AUROC (0.96) and AURAC (0.97), surpassing other well-performing models such as GPT-4 (entropy 0.9, AUROC 0.92, AURAC 0.90) and InternLM2 (entropy 1.5, AUROC 0.84, AURAC 0.85). This indicates that Chatlaw’s semantic entropy performance is stable, consistently delivering strong results across different question types and option formats. Our method also outperforms state-of-the-art models on open-ended questions, as shown in Table S2.

4.3. Real-world Legal Consultation

4.3.1. Performance Evaluation of Chatlaw

For the Lawbench benchmark, we utilize the official provided code. This test determines the final accuracy by analyzing the options in the large model's output and assessing their correctness. For judicial examination questions, we calculate the total scores for single-choice, multiple-choice, and uncertain multiple-choice questions based on actual objective questions from judicial exams and according to the official scoring standards. In real-world open-ended evaluation, we utilize four specific evaluation criteria:

Completeness: Can Chatlaw completely capture and present all important legal information in its response?

Correctness: Is Chatlaw's response accurate and consistent with established legal facts and principles?

Guidance: Does Chatlaw provide specific, practical, and actionable legal advice in its response?

Authority: Does Chatlaw's response cite relevant legal provisions, cases, or authoritative sources?

To conduct the evaluation, six levels of scoring were used: Strongly Agree, Agree, Neutral, Slightly Disagree, Disagree, and Strongly Disagree, corresponding to scores of 10, 8, 6, 4, 2, and 0, respectively. Legal experts were asked to evaluate Chatlaw's responses based on six predefined criteria and assign a score to each criterion. These scores allowed for a comprehensive assessment of Chatlaw's performance in the context of legal expertise and practical usability.

4.3.2. Performance on answering real-world legal consultation

In addition to evaluating accuracy and hallucinations based on objective questions, it is also crucial to assess how Chatlaw and other LLMs generate complete, informative, and authoritative consultation answers to real-world user inquiries (see the "Performance Evaluation of Chatlaw" section in the Methods). These consultation questions are based on online service records from law firms, covering a range of legal scenarios, including case consultations and legal advice. As with the objective evaluations, to better reflect the real-world application of a legal consultation AI assistant, each question was provided directly to the models without any task-specific or model-specific fine-tuning.

In Fig. 3, we present the expert evaluation results of Chatlaw and other LLMs on 250 open-ended questions. In the evaluation process, three legal experts independently assessed the answers from various AI legal assistants across four dimensions. The goal of the evaluation was to capture the multi-dimensional preferences of legal experts regarding the outputs of LLMs. Overall, we found that Chatlaw's answers ranked higher on average than those of other models tested, including GPT-4.

Chatlaw's average score is 8.71 (see Fig. 3(B)), which is 0.60 points higher than the second-ranked GPT-4. Compared to the open-source general-purpose model (the

best being InternLM2, 7.90) and the law-specific model (Fuzi-Mingcha, 7.24), Chatlaw’s improvements are even more substantial, with increases of 0.81 points and 1.47 points, respectively.

These results indicate that Chatlaw not only outperforms other models in terms of accuracy but also generates more desirable responses to a variety of legal consultation questions. Additionally, to better understand the relative strengths and weaknesses of different models, we analyzed their performance across different categories of questions.

5. Discussion

5.1. Principal Contributions and Implications

Overall, as shown in Fig. 1, our work establishes a comprehensive framework that begins with a unique legal data pipeline based on real-world law firm practices. This data foundation allowed us to train a novel RA-MoE model, which was then expanded into a complete multi-agent SOP framework. This architecture enables Chatlaw to function as a virtual law firm, providing end-to-end legal services. Crucially, our evaluations on two benchmarks and real-case consultations confirm that this specialized, role-aligned approach surpasses the capabilities of current powerful, general-purpose LLMs, including GPT-4.

5.2. Limitations and In-built Mitigation Strategies

Like all LLMs, our model is not immune to limitations such as hallucinations. In the legal field, this can manifest as fabricating non-existent statutes or citing outdated cases. However, our multi-agent paradigm is explicitly designed to mitigate this. We introduced a dedicated Legal Researcher agent, whose sole function is to retrieve and verify the latest legal provisions and relevant cases from external knowledge sources, thereby fact-checking and correcting the model’s outputs before they reach the user, as shown in Fig. 4. Similarly, to address robustness challenges (some users attempting to guide the model to incorrect answers through deception or concealment) our Legal Assistant agent employs knowledge graphs to guide users in providing all necessary information, ensuring the query is well-defined and grounded.

5.3. Deployment Challenges and Future Work

Our online trial phase also highlighted practical deployment challenges, particularly regarding privacy and data retention. Users are understandably divided: some consult on sensitive matters and demand no data be stored, while others wish to retain conversation history for long-term case support. Addressing this dichotomy is a key priority. We plan to strengthen the privacy protection architecture, enhancing communication and data storage security. Furthermore, we will explore various model compression techniques, such as knowledge distillation and quantization, to reduce the model’s computational footprint. Our ultimate goal is to develop a model sufficiently compact to run smoothly on users’ personal devices. This on-device deployment

paradigm would elegantly solve both the privacy concerns (as data never leaves the user’s device) and the computational resource issues, thereby enhancing accessibility and user trust.

5.4. Computational Overhead and Environmental Impact

A related deployment challenge is the demand for computational resources, particularly during inference. When a large number of users initiate requests simultaneously, it places extreme pressure on computational resources, which can lead to response delays. While our future work in model compression aims to solve this inference-time problem, it is also important to transparently report the one-time cost of training the current model. Our training procedure took place on a GPU server equipped with eight 80GB A100 GPUs and two Intel Xeon 8358P processors. The entire training process took approximately 23.14 hours, with an average power consumption of about 4.10 kilowatts. According to official statistics from Guangdong Province, the average carbon emission factor is 0.4512 kg of CO₂ per kWh consumed. Based on this, we estimate our model training process generated approximately 42.81 kilograms of additional CO₂ emissions. For high-efficient deployment, we also provide details on parallel acceleration on the Accelerating Deployment and Inference section in supplementary material.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] J.P. Heinz, *Urban lawyers: The new social structure of the bar*, University of Chicago Press, 2005.
- [2] J.S. Auerbach, *Unequal justice: Lawyers and social change in modern America*, Oxford University Press, New York, 1977.
- [3] M.V. Tushnet, *The rights revolution in the twentieth century*, American Historical Association, 2009.
- [4] T.B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, others, Language models are few-shot learners, in: *Advances in Neural Information Processing Systems*, 2020: pp. 1877–

- 1901.
- [5] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, G. Lample, LLaMA: Open and Efficient Foundation Language Models, (2023). <https://doi.org/10.48550/arXiv.2302.13971>.
 - [6] H.R. Kirk, B. Vidgen, P. Röttger, S.A. Hale, The benefits, risks and bounds of personalizing the alignment of large language models to individuals, *Nature Machine Intelligence* (2024) 1–10.
 - [7] J.W. Strachan, D. Albergo, G. Borghini, O. Pansardi, E. Scaliti, S. Gupta, K. Saxena, A. Rufo, S. Panzeri, G. Manzi, others, Testing theory of mind in large language models and humans, *Nature Human Behaviour* (2024) 1–11.
 - [8] R. Ren, S. Cai, X. Fang, X. Wang, Z. Zhang, M. Damiani, C. Hudlerova, A. Rosa, J. Hope, N.J. Cook, others, Multiplexed detection of viral antigen and RNA using nanopore sensing and encoded molecular probes, *Nature Communications* 14 (2023) 7362.
 - [9] L. Lv, Z. Lin, H. Li, Y. Liu, J. Cui, C.Y.-C. Chen, L. Yuan, Y. Tian, ProLLaMA: A Protein Large Language Model for Multi-Task Protein Language Processing, *arXiv Preprint arXiv:2402.16445* (2024).
 - [10] D. Peng, L. Zheng, D. Liu, C. Han, X. Wang, Y. Yang, L. Song, M. Zhao, Y. Wei, J. Li, others, Large-language models facilitate discovery of the molecular signatures regulating sleep and activity, *Nature Communications* 15 (2024) 3685.
 - [11] C. Kuenneth, R. Ramprasad, polyBERT: a chemical language model to enable fully machine-driven ultrafast polymer informatics, *Nature Communications* 14 (2023) 4099.
 - [12] N. Ferruz, S. Schmidt, B. Höcker, ProtGPT2 is a deep unsupervised language model for protein design, *Nature Communications* 13 (2022) 4348.
 - [13] A.J. Thirunavukarasu, D.S.J. Ting, K. Elangovan, L. Gutierrez, T.F. Tan, D.S.W. Ting, Large language models in medicine, *Nature Medicine* 29 (2023) 1930–1940.
 - [14] J. Clusmann, F.R. Kolbinger, H.S. Muti, Z.I. Carrero, J.-N. Eckardt, N.G. Laleh, C.M.L. Löffler, S.-C. Schwarzkopf, M. Unger, G.P. Veldhuizen, others, The future landscape of large language models in medicine, *Communications Medicine* 3 (2023) 141.
 - [15] S. Gilbert, H. Harvey, T. Melvin, E. Vollebregt, P. Wicks, Large language model AI chatbots require approval as medical devices, *Nature Medicine* 29 (2023) 2396–2398.
 - [16] C. Singh, A. Askari, R. Caruana, J. Gao, Augmenting interpretable models with large language models during training, *Nature Communications* 14 (2023) 7913.
 - [17] W. Fan, Y. Yang, J. Qi, Q. Zhang, C. Liao, L. Wen, S. Wang, G. Wang, Y. Xia, Q. Wu, others, A deep-learning-based framework for identifying and localizing multiple abnormalities and assessing cardiomegaly in chest X-ray, *Nature Communications* 15 (2024) 1347.
 - [18] D. Kiyasseh, A. Cohen, C. Jiang, N. Altieri, A framework for evaluating clinical artificial intelligence systems without ground-truth annotations, *Nature Communications* 15 (2024) 1808.

- [19] M. Nigo, L. Rasmy, B. Mao, B.S. Kannadath, Z. Xie, D. Zhi, Deep learning model for personalized prediction of positive MRSA culture using time-series electronic health records, *Nature Communications* 15 (2024) 2036.
- [20] S. Tayebi Arasteh, T. Han, M. Lotfinia, C. Kuhl, J.N. Kather, D. Truhn, S. Nebelung, Large language models streamline automated machine learning for clinical studies, *Nature Communications* 15 (2024) 1603.
- [21] OpenAI, GPT-4 Technical Report, (2023).
- [22] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, others, Llama: Open and efficient foundation language models, *arXiv Preprint arXiv:2302.13971* (2023).
- [23] J. Bai, S. Bai, Y. Chu, Z. Cui, K. Dang, X. Deng, Y. Fan, W. Ge, Y. Han, F. Huang, others, Qwen technical report, *arXiv Preprint arXiv:2309.16609* (2023).
- [24] Z. Du, Y. Qian, X. Liu, M. Ding, J. Qiu, Z. Yang, J. Tang, Glm: General language model pretraining with autoregressive blank infilling, *arXiv Preprint arXiv:2103.10360* (2021).
- [25] A. Zeng, X. Liu, Z. Du, Z. Wang, H. Lai, M. Ding, Z. Yang, Y. Xu, W. Zheng, X. Xia, others, Glm-130b: An open bilingual pre-trained model, *arXiv Preprint arXiv:2210.02414* (2022).
- [26] I. Team, InternLM: A Multilingual Language Model with Progressively Enhanced Capabilities, (2023). <https://github.com/InternLM/InternLM>.
- [27] Z. Cai, M. Cao, H. Chen, K. Chen, K. Chen, X. Chen, X. Chen, Z. Chen, Z. Chen, P. Chu, others, Internlm2 technical report, *arXiv Preprint arXiv:2403.17297* (2024).
- [28] W. Deng, J. Pei, K. Kong, Z. Chen, F. Wei, Y. Li, Z. Ren, Z. Chen, P. Ren, Syllogistic Reasoning for Legal Judgment Analysis, in: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023: pp. 13997–14009.
- [29] H.-T. Nguyen, A Brief Report on LawGPT 1.0: A Virtual Legal Assistant Based on GPT-3, *arXiv Preprint arXiv:2302.05729* (2023).
- [30] Q. Huang, M. Tao, Z. An, C. Zhang, C. Jiang, Z. Chen, Z. Wu, Y. Feng, Lawyer LLaMA Technical Report, *ArXiv abs/2305.15062* (2023).
- [31] E. Mik, Caveat Lector: Large Language Models in Legal Practice, *arXiv Preprint arXiv:2403.09163* (2024).
- [32] R.T. McCoy, S. Yao, D. Friedman, M. Hardy, T.L. Griffiths, Embers of autoregression: Understanding large language models through the problem they are trained to solve, *arXiv Preprint arXiv:2309.13638* (2023).
- [33] J. Maynez, S. Narayan, B. Bohnet, R. McDonald, On faithfulness and factuality in abstractive summarization, *arXiv Preprint arXiv:2005.00661* (2020).
- [34] D. Li, Y. Ma, N. Wang, Z. Ye, Z. Cheng, Y. Tang, Y. Zhang, L. Duan, J. Zuo, C. Yang, others, Mixlor: Enhancing large language models fine-tuning with lora-based mixture of experts, *arXiv Preprint arXiv:2404.15159* (2024).
- [35] Y. Hua, H. Cao, Z. Tao, B. Li, Z. Wu, C. Liu, L. Xu, Input Domain Aware MoE: Decoupling Routing Decisions from Task Optimization in Mixture of Experts, *arXiv Preprint arXiv:2510.16448* (2025).
- [36] D.A. Boiko, R. MacKnight, B. Kline, G. Gomes, Autonomous chemical research

- with large language models, *Nature* 624 (2023) 570–578.
- [37] D. Lepikhin, H.-J. Loh, W. Fedus, Y. Chen, others, Gshard: Scaling giant models with conditional computation and automatic sharding, in: *ICLR*, 2020.
 - [38] W. Fedus, B. Zoph, N. Shazeer, Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, *Journal of Machine Learning Research* 23 (2022) 1–39.
 - [39] A. Liu, B. Feng, B. Xue, B. Wang, B. Wu, C. Lu, C. Zhao, C. Deng, C. Zhang, C. Ruan, others, Deepseek-v3 technical report, *arXiv Preprint arXiv:2412.19437* (2024).
 - [40] A.Q. Jiang, A. Sablayrolles, A. Roux, A. Mensch, B. Savary, C. Bamford, D.S. Chaplot, D. de las Casas, E.B. Hanna, F. Bressand, others, Mixtral of experts, *arXiv Preprint arXiv:2401.04088* (2024).
 - [41] J. Krajewski, J. Ludziejewski, K. Adamczewski, M. Pióro, M. Krutul, S. Antoniak, K. Ciebiera, K. Król, T. Odrzygóźdź, P. Sankowski, others, Scaling laws for fine-grained mixture of experts, *arXiv Preprint arXiv:2402.07871* (2024).
 - [42] S. Hong, M. Zhuge, J. Chen, X. Zheng, Y. Cheng, J. Wang, C. Zhang, Z. Wang, S.K.S. Yau, Z. Lin, others, MetaGPT: Meta programming for a multi-agent collaborative framework, in: *The Twelfth International Conference on Learning Representations*, 2023.
 - [43] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K.R. Narasimhan, Y. Cao, React: Synergizing reasoning and acting in language models, in: *The Eleventh International Conference on Learning Representations*, 2022.
 - [44] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu, others, Autogen: Enabling next-gen LLM applications via multi-agent conversations, in: *First Conference on Language Modeling*, 2024.
 - [45] C. Pei, Z. Wang, F. Liu, Z. Li, Y. Liu, X. He, R. Kang, T. Zhang, J. Chen, J. Li, others, Flow-of-Action: SOP Enhanced LLM-Based Multi-Agent System for Root Cause Analysis, in: *Companion Proceedings of the ACM on Web Conference 2025*, 2025: pp. 422–431.
 - [46] B. Gan, Y. Zhao, T. Zhang, J. Huang, Y. Li, S.X. Teo, C. Zhang, W. Shi, MASTER: A Multi-Agent System with LLM Specialized MCTS, *arXiv Preprint arXiv:2501.14304* (2025).
 - [47] J.M. Niebla Zatarain, Artificial intelligence and legal analytics: New tools for law practice in the digital age, (2018).
 - [48] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, H. Wang, H. Wang, Retrieval-augmented generation for large language models: A survey, *arXiv Preprint arXiv:2312.10997 2* (2023).
 - [49] Z. Fei, X. Shen, D. Zhu, F. Zhou, Z. Han, A. Huang, S. Zhang, K. Chen, Z. Yin, Z. Shen, others, Lawbench: Benchmarking legal knowledge of large language models, in: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024: pp. 7933–7962.
 - [50] I. Chalkidis, A. Jana, D. Hartung, M. Bommarito, I. Androutsopoulos, D.M. Katz, N. Aletras, LexGLUE: A benchmark dataset for legal language understanding in English, *arXiv Preprint arXiv:2110.00976* (2021).

- [51]Z. Zhou, J.-X. Shi, P.-X. Song, X.-W. Yang, Y.-X. Jin, L.-Z. Guo, Y.-F. Li, Lawgpt: A chinese legal knowledge-enhanced large language model, arXiv Preprint arXiv:2406.04614 (2024).
- [52]S. Wu, Z. Liu, Z. Zhang, Z. Chen, W. Deng, W. Zhang, J. Yang, Z. Yao, Y. Lyu, X. Xin, S. Gao, P. Ren, Z. Ren, Z. Chen, fuzi.mingcha, GitHub Repository (2023). <https://github.com/irlab-sdu/fuzi.mingcha>.
- [53]S. Dhuliawala, M. Komeili, J. Xu, R. Raileanu, X. Li, A. Celikyilmaz, J. Weston, Chain-of-verification reduces hallucination in large language models, arXiv Preprint arXiv:2309.11495 (2023).
- [54]S. Farquhar, J. Kossen, L. Kuhn, Y. Gal, Detecting hallucinations in large language models using semantic entropy, Nature 630 (2024) 625–630.
- [55]S. Lin, J. Hilton, O. Evans, Teaching models to express their uncertainty in words, arXiv Preprint arXiv:2205.14334 (2022).
- [56]S. Lin, J. Hilton, O. Evans, Truthfulqa: Measuring how models mimic human falsehoods, arXiv Preprint arXiv:2109.07958 (2021).

Figures

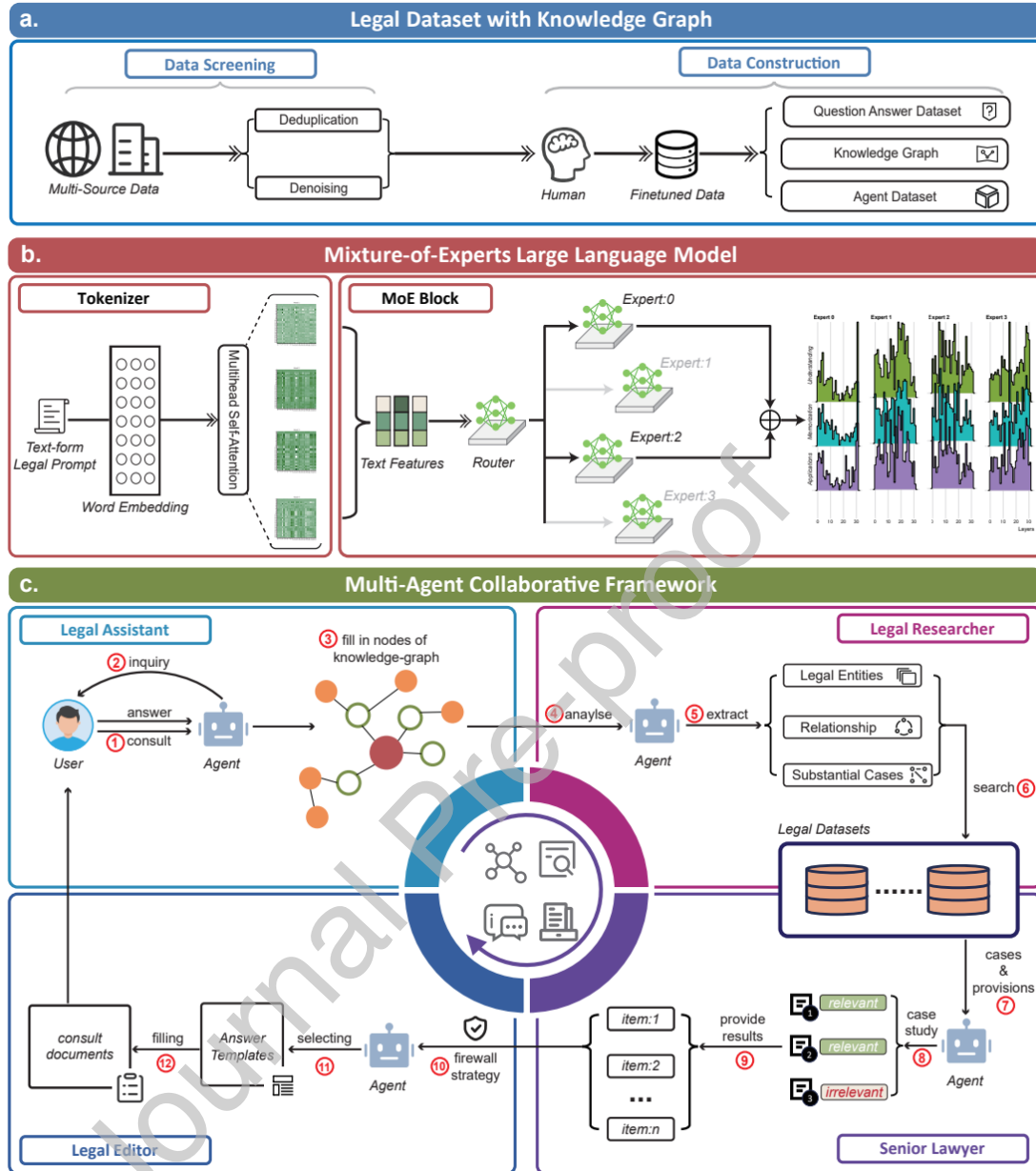


Fig. 1. The Framework of Chatlaw. Our framework consists of three core components: **(a) Data Processing Pipeline:** We screen and structure multi-source data to construct a high-quality legal dataset enriched with a knowledge graph. **(b) Mixture-of-Experts (MoE) Model:** The core language model utilizes an MoE architecture, where a router directs legal prompts to specialized expert sub-networks for efficient and accurate processing. **(c) Multi-Agent Collaborative Framework:** A system of four specialized agents (**Legal Assistant**, **Researcher**, **Editor**, and **Senior Lawyer**) that simulates a legal team's workflow. The framework handles the entire consultation process, from initial user inquiry and legal research to document editing and final review.

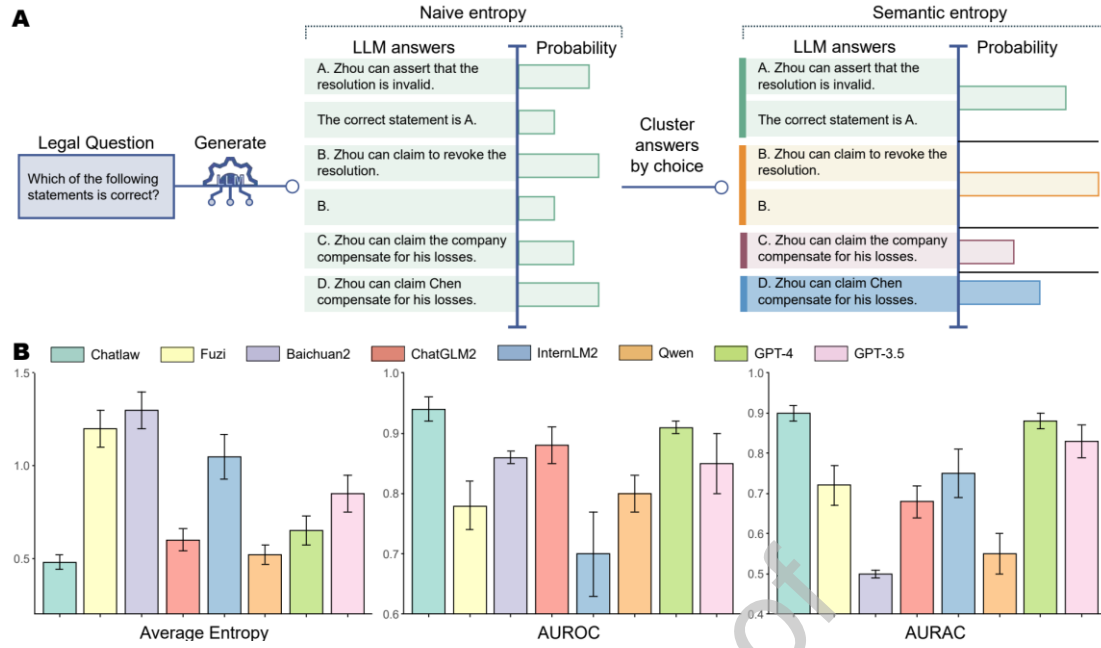


Fig. 2. Naive vs. semantic entropy in legal question answering. (A) Illustration of naive entropy (probability distribution over raw outputs) and semantic entropy (clustering by meaning before entropy calculation). (B) Comparison of LLMs on average entropy, AUROC, and AURAC. Chatlaw shows lower entropy and higher accuracy, indicating more reliable performance. Error bars denote standard deviation.

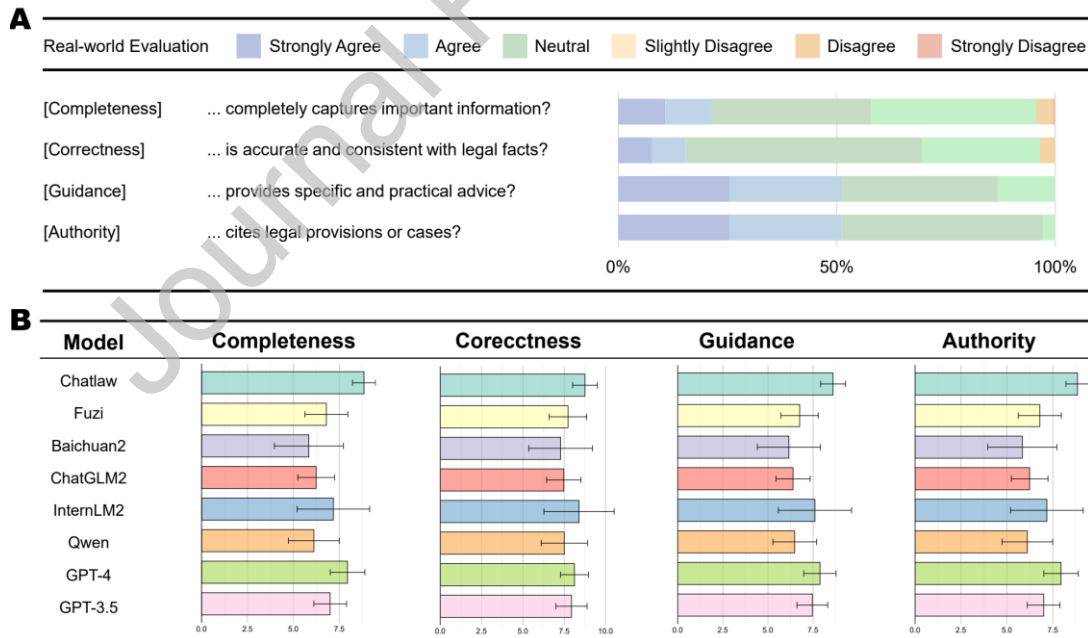


Fig. 3. Performance Evaluation on real-world legal consultation. (A) The criteria for assessing legal consultation quality, including Completeness, Correctness, Guidance, and Authority, rated on a scale from 0 to 10. Additionally, the right-side stacked bar chart presents the distribution of Chatlaw's responses across all cases, categorized into six agreement levels. (B) The performance comparison of various

models (GPT-4, GPT-3.5, Baichuan2, ChatGLM2, InternLM2, Qwen, Fuzi, and Chatlaw) across these criteria, with Chatlaw consistently achieving the highest scores. Error bars represent standard deviation (std).

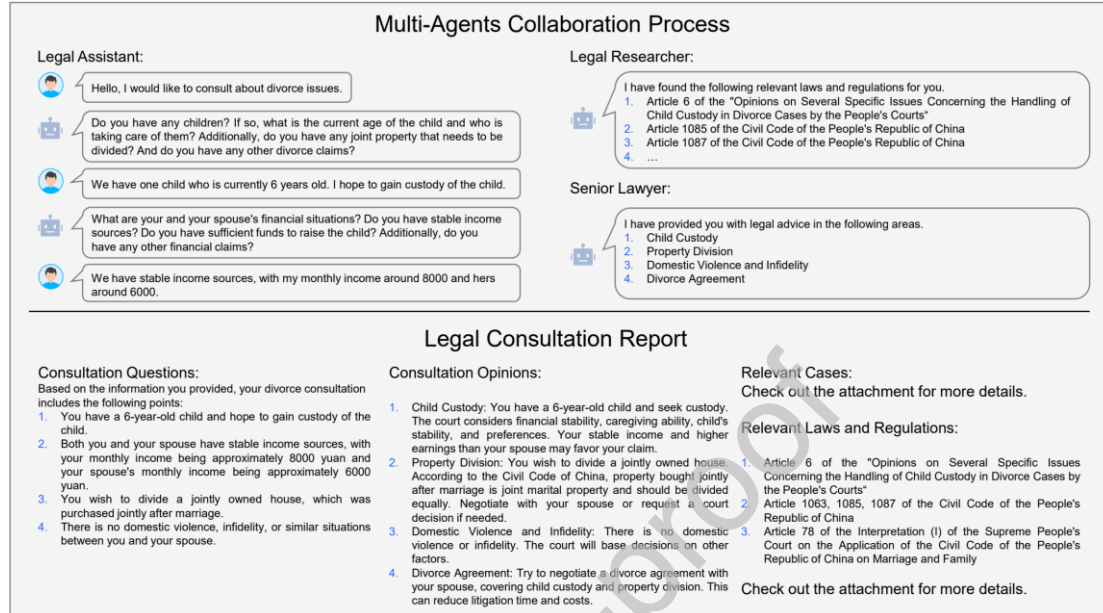


Fig. 4. Multi-Agent Collaboration Process and Legal Consultation Report. The figure illustrates the collaboration workflow among three specialized agents in a divorce case consultation. The process begins with the **Legal Assistant**, who interacts with the user to gather essential facts (e.g., children, finances). Subsequently, the **Legal Researcher** retrieves pertinent legal articles and regulations. Finally, the **Senior Lawyer** synthesizes this information to deliver structured advice across key areas like child custody and property division. This process culminates in a comprehensive Legal Consultation Report, which summarizes the user's circumstances into Consultation Questions, provides actionable Consultation Opinions, and cites the supporting Relevant Laws and Regulations.

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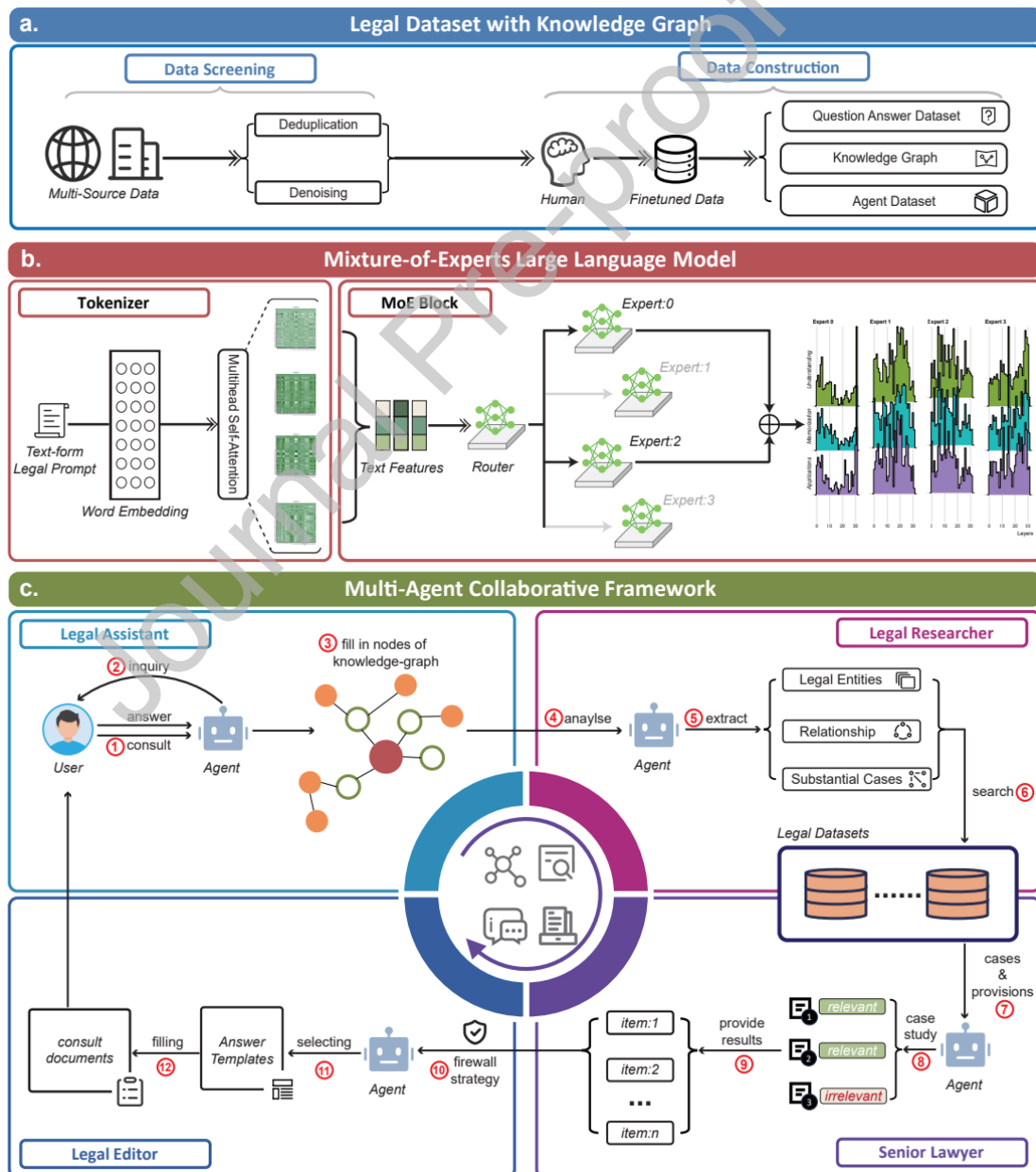
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Graphical Abstract

Chatlaw is an advanced AI legal assistant that improves the accuracy and reliability of legal services. It integrates a high-quality legal dataset, a Role-Aligned Mixture-of-Experts (RA-MoE) architecture, and a multi-agent system to provide precise, case-specific legal advice. By emulating real law firm workflows, it reduces errors and outperforms general AI models like GPT-4, achieving better results in legal benchmarks and real-world evaluations. Chatlaw makes legal services more accessible, marking a significant step forward in AI for law.



Journal Pre-proof

Declaration of Interest Statement

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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